

ATHARVA HATEKAR

+49 1575 8905132 | atharva.hatekar@outlook.com | Frankfurt, Hesse, Germany (Available for relocation)

Portfolio: atharvahatekar.github.io | linkedin.com/in/atharvahatekar | github.com/atharvahatekar

SUMMARY

Data Scientist / AI Engineer with 4+ years of experience designing, scaling, and deploying production-grade AI systems. Engineered real-time anomaly detection and forecasting models across 100+ industrial sensors, **improving predictive maintenance accuracy by 40%**, and customized a Graph Deviation Network with multi-head attention to capture 17 true anomalies missed by baseline thresholds. Developed advanced generative AI architectures, including a medical-SLM from scratch in PyTorch and a guardrailed agentic RAG pipeline with automated evaluation. Drives the full MLOps lifecycle across AWS and Azure to deliver resilient, continuously monitored enterprise AI solutions.

TECHNICAL SKILLS

Languages & Backend: Python, SQL, Bash, FastAPI, Pydantic, REST APIs, gRPC, PostgreSQL, Redis

Machine Learning & Data Science: Scikit-learn, XGBoost, PyTorch, Pandas, NumPy, Statistics, Statistical Modeling, Feature Engineering, Model Evaluation, Time-Series Analysis & Forecasting, Anomaly Detection, NLP

AI Engineering & GenAI: LLMs/SLMs, Agentic AI, RAG, Information Retrieval, LangChain, LangGraph, Fine-Tuning (LoRA/QLoRA), Quantization, Transformers, HuggingFace, Vector Databases (Qdrant, FAISS), vLLM, Groq, RAGAS, LangSmith, LLM Evaluation, AI Guardrails

MLOps & Infrastructure: Git, Docker, Kubernetes, MLflow, Weights & Biases, Apache Airflow, Great Expectations, Kafka, PySpark, GitHub Actions, CI/CD, Pytest, Terraform, Model Deployment, Drift Monitoring & Observability

Cloud Platforms: AWS (SageMaker, Bedrock, Lambda, S3, EC2, ECR, CloudWatch), Azure (Azure ML, Azure DevOps)

Visualization & Analytics: Plotly, Grafana, Streamlit, Power BI, Data Visualization

WORK EXPERIENCE

Metso, Oberursel, Germany 10/2024 – 06/2026

Data Scientist (Thesis Student) 11/2025 – 06/2026

- Enhanced a Graph Deviation Network with multi-head attention and prior-informed structures, **capturing 17 critical anomalies missed by baseline thresholds** after benchmarking against streaming-PCA and Adaptive-PCA models.
- Architected Dockerized training and inference pipelines with CI/CD on Azure, **scaling processing capacity to transform raw live feeds from 500+ sensor tags** into model-ready features.
- Shipped an enterprise-grade Grafana root-cause dashboard, **slashing diagnostic time for process engineers** by enabling real-time alert tracing to specific sensor groups during live equipment testing.

Data Scientist (Working Student)

04/2025 – 10/2025

- Designed and scaled a real-time monitoring system processing **100+ live process signals**, integrating advanced outlier detection (Isolation Forest, Autoencoders) and forecasting models (ARIMA, Prophet, LSTM, Transformers) to capture early drifts and faults.
- Partnered with plant experts to cross-validate model predictions against physical asset behavior, translating domain expertise into custom feature engineering that **improved predictive maintenance accuracy by 40%**.

Data Science Intern

10/2024 – 03/2025

- Integrated ACT process control software via **gRPC** to ingest live data streams, developing the plant's first **Grafana KPI dashboard** that combined linear regression sensor trends with domain-informed health metrics to centralize equipment condition tracking for engineering teams.

Dreamwarez, Pune, India

02/2021 – 02/2023

Data Analyst

08/2021 – 02/2023

- Engineered end-to-end Python and SQL data pipelines from scratch to consolidate multi-source hospitality data, identifying revenue leakage and driving a migration to seasonal dynamic pricing that **increased total revenue by 23%**.
- Built and cross-validated XGBoost and Linear Regression demand forecasts across 200+ properties, achieving 16% MAPE and driving rate-structure changes that lifted RevPAR by 4.4%.

- Developed centralized Power BI dashboards leveraging optimized DAX metrics to unify infrastructure data, completely automating executive reporting and **saving 12+ hours of manual spreadsheet workflows weekly**.

Software Developer Intern02/2021 – 07/2021

- Optimized an internal ticketing platform by refactoring backend pipelines and implementing robust input validation, which significantly reduced systemic bugs and system downtime.

SELECTED PROJECTS

Medical-SLM | Link06/2026 – 08/2026

- Engineered a **60M-parameter Transformer-based Medical Small Language Model (SLM)** from scratch in PyTorch, hand-coding high-performance components (multi-head latent attention, RoPE, RMSNorm, SwiGLU) and optimizing pre-training on **600M+** tokens under strict consumer GPU constraints.
- Orchestrated the complete model lifecycle independently, implementing streaming Parquet data ingestion, custom tokenizer training, and instruction tuning to deliver a **specialized zero-shot evaluation harness** for MedMCQA and PubMedQA benchmarks.

Enterprise Agentic RAG | Link06/2026 – 07/2026

- Developed a production-ready, multi-agent RAG platform utilizing **LangGraph** and **FastAPI** backed by **Qdrant** and **Groq** inference, implementing stateful conversation memory and multi-step reasoning to automate complex querying over enterprise document datasets.
- Engineered a secure retrieval pipeline featuring semantic chunking, FlashRank reranking, and history-aware query planning, **slashing hallucination risks and expanding contextual grounding**.
- Integrated advanced AI Guardrails to ensure robust jailbreak prevention while embedding Logfire, LangSmith, and RAGAS to achieve complete end-to-end tracing and automated evaluation workflows.

Customer Churn Prediction MLOps | Link06/2026 – 07/2026

- Engineered and deployed an end-to-end **XGBoost churn prediction model on AWS**, integrating **Great Expectations** for automated input data validation, **MLflow** for experiment tracking, and **GitHub Actions** for robust CI/CD automation.
- Containerized the architecture utilizing **Docker** and **FastAPI** to serve high-throughput inference endpoints, establishing real-time model reliability and data drift monitoring to ensure long-term production stability.

Hybrid Log Classification System | Link01/2026 – 02/2026

- Designed an IT operations log triage service that routes high-volume system messages through a cost-tiered architecture, leveraging regex rules for predictable patterns, Sentence Transformer embeddings with Logistic Regression for complex records, and LLM fallbacks for zero-shot sparse data scenarios.
- Achieved **99% accuracy across 2,400+ infrastructure logs** (security alerts, critical errors, resource metrics), containerizing the system via FastAPI to fully automate live incident labeling and ticket routing.

EDUCATION

Brandenburg University of Technology, Cottbus, Germany04/2023 – 08/2026

Master of Science in Artificial Intelligence | Grade: 1.7 (Very Good)

Thesis (Grade: 1.0 / Excellent): Unsupervised anomaly detection on non-stationary industrial multivariate time-series.

Lovely Professional University, Jalandhar, India06/2017 – 06/2021

B.Tech. in Computer Science and Engineering | Grade: 2.0 (Good)

PUBLICATION

Passenger Security Using Facial Emotion Recognition System | Link09/2022 – 11/2022

LANGUAGES

English (C1, Professional) | German (B1, Intermediate) | Hindi (Native) | Marathi (Native)

ACHIEVEMENTS

- Secured All India Rank 2,757 out of 157,000+ candidates in GATE (Graduate Aptitude Test in Engineering), placing in the **Top 1.76%** (98.24th percentile) nationwide.